

Python Automation

Project Description

Applied basic Python scripting to remove unauthorized IP addresses from "allow_list.txt" and automate a routine access control task.

```
# Removes unauthorized IPs and creates new allowed list

def update_file(import_file, remove_list):
    with open(import_file, "r") as file:
        ip_addresses = file.read().split()

    # Remove unauthorized IPs
    ip_addresses = [ip for ip in ip_addresses if ip not in remove_list]

    # Write updated list back to file
    with open(import_file, "w") as file:
        file.write("\n".join(ip_addresses))

# Example usage

update_file(
    "allow_list.txt",
    ["192.168.6.9", "192.168.52.90"]
)
```

Script Logic

The script reads IP addresses from a file, converts them into a list, and filters out any entries that appear in the removal list. The updated list is then written back to the file.

Process Steps

- Read IP addresses from a file "allow_list.txt"
- Convert file contents into ip_addresses list for processing
- Compare entries against removal list remove_list
- Filter out unauthorized IP addresses using list comprehension
- Write updated list back to file "allow_list.txt"

Functions

- File handling (`with open`, read/write modes)
- Data transformation (`.read()`, `.split()`, `.join()`)
- Iteration and conditional filtering (`for`, `if`)

Summary

I used a Python script to automate a basic access control task, removing unauthorized IP addresses from an allow list. This demonstrates how scripting can be applied to support simple security operations, reduce manual updates, and maintain consistency in access control configurations.